SECSR2043 OPERATING SYSTEMS
[20 Marks]

Name	:	NURUL ADRIANA BINTI KAMAL JEFRI	Marks
Student ID	:	A23CS0258	
Section	:	03	

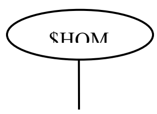
Instruction: Please answer all of the following questions. Whenever the ☝ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

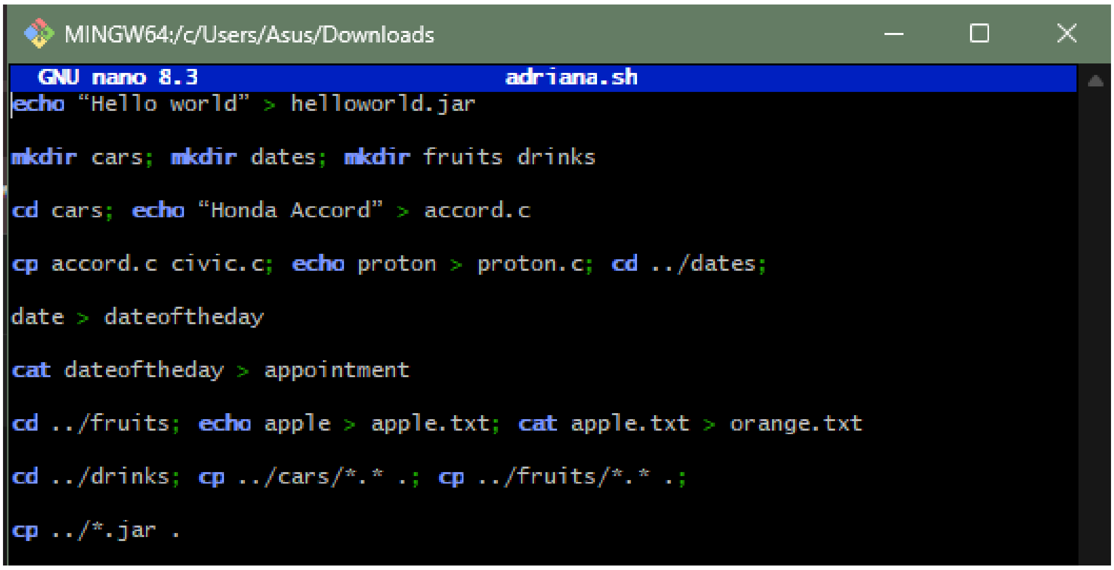
1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo “Hello world” > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo “Honda Accord” > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begins with **$HOME** directory.

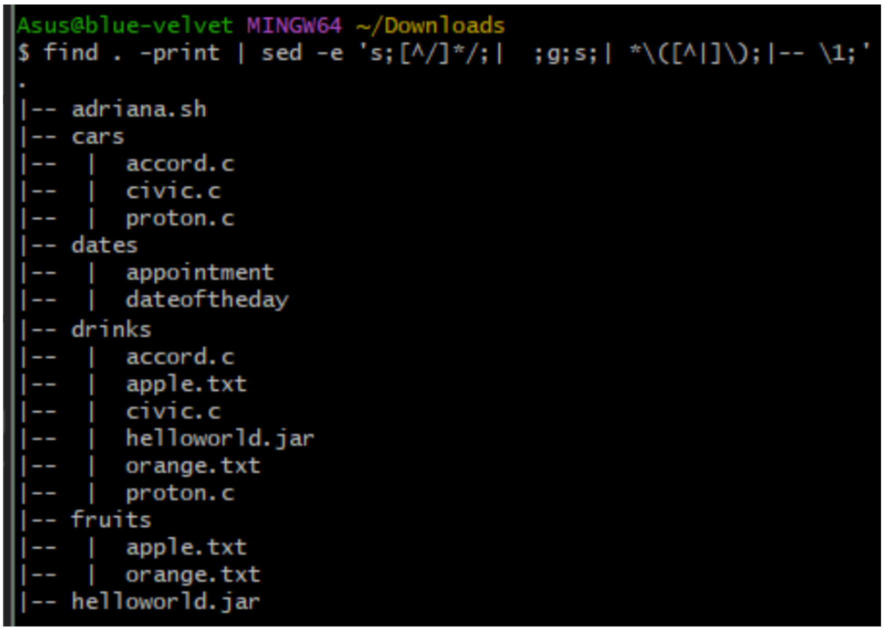
[4 marks]





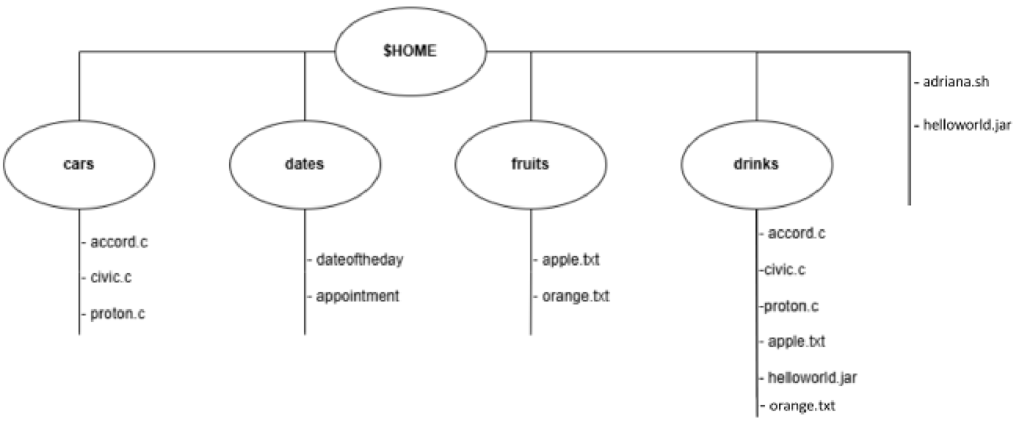
```
MINGW64:/c/Users/Asus/Downloads
GNU nano 8.3                    adriana.sh
echo "Hello world" > helloworld.jar

mkdir cars; mkdir dates; mkdir fruits drinks

cd cars; echo "Honda Accord" > accord.c

cp accord.c civic.c; echo proton > proton.c; cd ../dates;

date > dateoftheday

cat dateoftheday > appointment

cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt

cd ../drinks; cp ../cars/*.* .; cp ../fruits/*.* .;

cp ../*.jar .
```

Full script of adriana.sh file



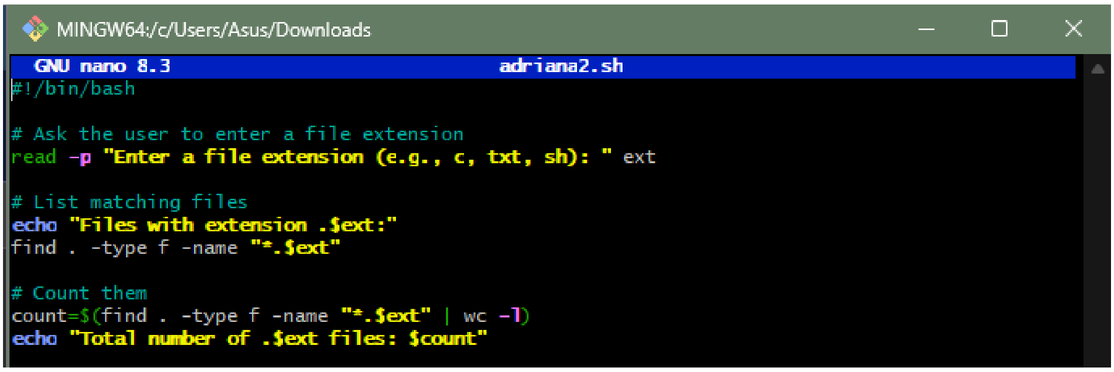
```
Asus@blue-velvet MINGW64 ~/Downloads
$ find . -print | sed -e 's;[^/]*/;|  ;g;s;| *\([^|]\);|-- \1;'
.
|-- adriana.sh
|-- cars
|--  |  accord.c
|--  |  civic.c
|--  |  proton.c
|-- dates
|--  |  appointment
|--  |  dateoftheday
|-- drinks
|--  |  accord.c
|--  |  apple.txt
|--  |  civic.c
|--  |  helloworld.jar
|--  |  orange.txt
|--  |  proton.c
|-- fruits
|--  |  apple.txt
|--  |  orange.txt
|-- helloworld.jar
```

Tree structure in Downloads folder

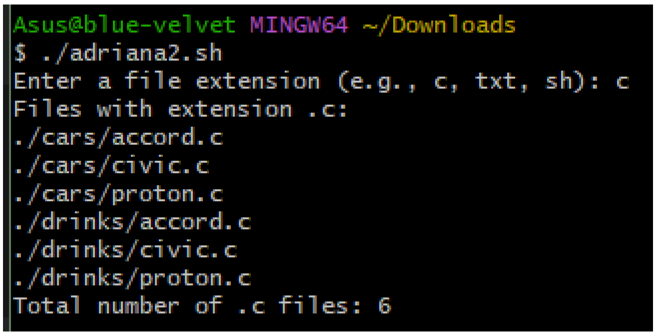


Full directory tree structure

b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display `c` program files, and enter "`c`" as the input to the bash script. [4 marks]



Full script of adriana2.sh file



Output after running the script with input `c`

2. The following Figure 1 illustrates a tree structure of some directories and files.

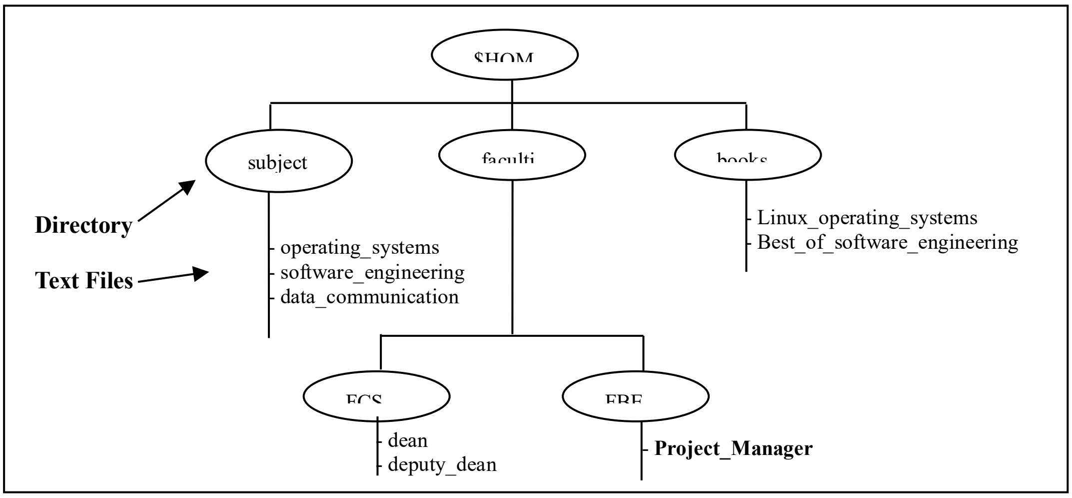


Figure 1

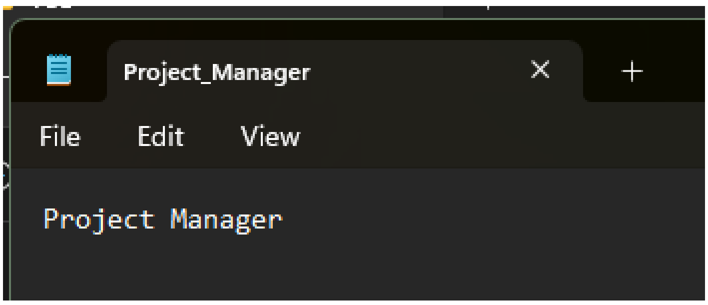
a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

```
MINGW64:/c/Users/Asus/Downloads
  GNU nano 8.3                              myname2a.sh
#!/bin/bash

# Create main directories
mkdir -p subjects/{operating_systems,software_engineering,data_communication}
mkdir -p faculties/{FCS,FBE}
mkdir -p books

# Create text files under 'books' directory
echo "Linux operating systems" > books/Linux_operating_systems
echo "Best of software engineering" > books/Best_of_software_engineering

# Create text files inside correct faculty folders
echo "dean" > faculties/FCS/dean
echo "deputy dean" > faculties/FCS/deputy_dean
echo "Project Manager" > faculties/FBE/Project_Manager

echo "Script executed successfully!"
```

Full script of myname2a.sh file

```
Asus@blue-velvet MINGW64 ~/Downloads
$ find . -print | sed -e 's;[^/]*/;|  ;g;s;| *\([^|]\);|-- \1;'
.
|-- books
|--  |  Best_of_software_engineering
|--  |  Linux_operating_systems
|-- faculties
|--  |  FBE
|--  |  |  Project_Manager
|--  |  FCS
|--  |  |  dean
|--  |  |  deputy_dean
|-- myname2a.sh
|-- subjects
|--  |  data_communication
|--  |  operating_systems
|--  |  software_engineering
```

Folder view of myname2a.sh file



Project Manager inside Project_Manager file

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
books	drwxrw-xr-x
subjects	**drwxr-xr-x**
Best_of_software_engineering	**-rw-r--r--**
FCS	**drwxr-xr-x**
project_manager	**-rw-r--r--**

```
Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ld books
drwxr-xr-x 1 Asus 197121 0 Jun 17 03:09 books

Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ld subjects
drwxr-xr-x 1 Asus 197121 0 Jun 17 03:09 subjects
Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ls books
total 2
1 -rw-r--r-- 1 Asus 197121 29 Jun 17 03:19 Best_of_software_engineering
Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ld faculties/FCS
drwxr-xr-x 1 Asus 197121 0 Jun 17 03:09 faculties/FCS

Asus@blue-velvet MINGW64 ~/Downloads
$ ls -l faculties/FBE/Project_Manager
-rw-r--r-- 1 Asus 197121 16 Jun 17 03:19 faculties/FBE/Project_Manager
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

```
MINGW64:/c/Users/Asus/Downloads
  GNU nano 8.3                    myname2c.sh
#!/bin/bash

# subjects: Owner and Group execute
chmod 770 subjects

# Best_of_software_engineering: all can execute
chmod 775 books/Best_of_software_engineering

# FCS: all can execute
chmod 775 faculties/FCS

# project_manager: all can execute
chmod 775 faculties/FBE/Project_Manager

echo "Permissions updated!"
```

Full script of myname2c.sh file

```
Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ld subjects
drwxrwx--- 1 Asus 197121 0 Jun 17 03:09 subjects

Asus@blue-velvet MINGW64 ~/Downloads
$ ls -l books/Best_of_software_engineering
-rwxrwxr-x 1 Asus 197121 29 Jun 17 03:19 books/Best_of_software_engineering

Asus@blue-velvet MINGW64 ~/Downloads
$ ls -ld faculties/FCS
drwxrwxr-x 1 Asus 197121 0 Jun 17 03:09 faculties/FCS

Asus@blue-velvet MINGW64 ~/Downloads
$ ls -l faculties/FBE/Project_Manager
-rwxrwxr-x 1 Asus 197121 16 Jun 17 03:19 faculties/FBE/Project_Manager
```

New permissions after running myname2c.sh file

d) Complete the **following** table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	**drwxrwx---**
Best_of_software_engineering	**-rwxrwxr-x**
FCS	**drwxrwxr-x**
project_manager	**-rwxrwxr-x**

End of Lab 3

***** All the Best for Final Exam *****